

Summary

AI engineer with experience building LLM-powered applications, retrieval systems, and production backend infrastructure. Worked across tool-calling models, document intelligence, RAG pipelines, and cloud-native services, with hands-on experience in model training, evaluation, and deployment.

Education

B.Tech in Chemical Engineering	Indian Institute of Technology Madras	7.64	2023–2027
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Achievements

- **98.61 percentile** in JEE Mains (1.2M+ candidates); **Top 450** in WBJEE; **AIR 71** (Junior Squad) and **City Topper** in Technothon Prelims.

Relevant Coursework

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|---|---------------------------------------|--|
| • Full-Stack Development (React, Node.js) | • Mathematics for ML (Linear Algebra) | • Deep Learning Specialization (Andrew Ng) |
| • Practical Deep Learning (fast.ai) | • LLM Fine-Tuning & Optimization | • Machine Learning in Production |
| • Generative AI for LLMs | | |

Technical Skills

Languages	Python, C++, JavaScript, Bash
LLM & ML	PyTorch, Transformers, QLoRA, LoRA, PEFT, Instruction Tuning, Tool Calling, Function Calling
Retrieval	LangChain, FAISS, ChromaDB, Semantic Search, Hybrid Retrieval, Reranking, Chunking Strategies
Document AI	Multimodal AI, OCR, PaddleOCR, YOLOv10, CV Layout Detection, ODParse
Backend	FastAPI, REST APIs, AsyncIO, PostgreSQL, Supabase, asyncpg
Data & Pipelines	Airflow, Apify, NumPy, Pandas, BeautifulSoup
Cloud & Infra	Docker, GCP, Azure, Modal, Railway, CI/CD, Linux
Tools	Git, Cursor, Claude Code, GitHub Copilot

Experience

Otsuka Corporation

Tokyo, Japan

Software Development Engineer Intern

May 2026 – Jul. 2026

- Developed scalable synthetic data generation pipelines producing **50K+** function-calling examples spanning multi-turn reasoning, execution-grounded supervision, behavioral alignment, and parallel tool invocation for open-source LLM training.
- Built reproducible **QLoRA** fine-tuning and automated evaluation pipelines for **Qwen3-8B**, achieving **97%** BFCL accuracy on single-turn function-calling tasks while designing trajectory-level behavioral diagnostics and benchmarking infrastructure on **NVIDIA DGX Spark**.
- Architected **Senpai**, an enterprise AI execution platform that transforms natural-language objectives into dependency-aware execution DAGs through planner-driven capability graphs, adaptive scheduling, runtime DAG expansion, GraphRAG, and evidence-grounded reasoning.
- Optimized production LLM inference using Atlas, prefix caching, persistent context caching, **SSM cache-slot reuse**, and parallel execution, increasing generation throughput by **5× (11 → 55 tokens/sec)** while reducing end-to-end response latency for production AI workloads.
- Engineered production-grade agent runtimes with deterministic-first orchestration, intelligent tool routing, streaming execution, hybrid retrieval, and modular capability composition to improve scalability, reliability, and enterprise deployment readiness.

Gravton Labs

Ontario, Canada (Remote)

AI Engineer Intern

Feb. 2026 – May 2026

- Architected GEO retrieval pipelines across Reddit and Quora using SERP-driven discovery, Apify-based extraction, and Airflow DAG orchestration for AI visibility intelligence at scale.
- Designed hybrid ingestion architectures combining actor-backed cold-start discovery with stable extraction pipelines, improving retrieval coverage for buyer-intent and comparison-intent queries.
- Migrated backend from Encore TypeScript to **FastAPI** microservices deployed on Railway with Supabase, improving maintainability and enabling Python-native async workflows
- Built async enrichment workflows using Airflow, asyncpg, and cloud-native orchestration; added structured telemetry and observability to surface pipeline failures and retrieval coverage gaps in production.
- Developed citation attribution systems and AI visibility metrics powering Gravton’s Insights Engine for competitor intelligence and brand monitoring across LLM-generated content.

OctonData

San Francisco, CA (Remote)

Software Engineer Intern

Oct. 2025 – Dec. 2025

- Primary engineer for an end-to-end document intelligence platform processing **10K+ pages/month** across ingestion, parsing, APIs, and developer workflows.
- Architected a hybrid document parsing pipeline combining OCR, CV-based layout detection, multimodal encoders, and LLM fallback routing, improving robustness on complex and variable tax document formats.
- Designed cloud-native workflows on **GCP** for secure ingestion, local processing, and structured output generation under compliance constraints; implemented idempotent reprocessing and duplicate detection across multi-stage pipelines.
- Built scalable REST APIs (`/parse`, `/extract`) with authenticated access, rate limiting, and usage metering; developed a self-serve developer portal with parsing previews and credit-based access control.
- Improved long-document RAG accuracy by **20–30%** using semantic, recursive, and hybrid chunking strategies; contributed deep-learning modules for layout detection, adaptive segmentation, and table understanding.
- Deployed production Tax CPA Parser; onboarded **10+ U.S. CPA firms** and reduced manual document review time by **40%+**.

MACHINE LEARNING INTERN

Sep. 2025 – Oct. 2025

- Built multilingual document parsing pipelines supporting **5+ languages** using YOLOv10, PP-DocLayout-L, and PaddleOCR with RTL and Devanagari script support.
- Designed ensemble-based layout inference with rotation handling and adaptive scaling, improving parsing robustness across varied document types and scan qualities.
- Integrated **Qwen3-VL** for multimodal understanding of tables, figures, and charts; embedded outputs via Gemma for ChromaDB indexing and downstream retrieval.
- Ranked **Top 20 nationally** for solo pipeline contribution; Tecnod8 recognized in *Forbes India Select 200* during tenure.

Projects

MedProQA – Fine-tuned LLM for Medical Question Answering

Kaggle

SOLO DEVELOPER

Jul. 2025

- Fine-tuned **Phi-3** using QLoRA with 4-bit quantization, reducing GPU memory usage by **38%** and training cost by **24%** relative to full fine-tuning baseline.
- Curated and preprocessed **250K+ medical QA pairs** from MedQA, MedMCQA, and PubMedQA with schema normalization and deduplication pipelines.
- Achieved **82.6% accuracy on MedMCQA** benchmark; reduced hallucination rate by **28%** on held-out out-of-distribution queries, measured via factual consistency scoring against source context.

R.E.A.C.H. – AI-Powered Emergency Response Platform

IITM Nirmaan Cohort

CO-FOUNDER & AI LEAD

Apr. 2025 – Present

- Fine-tuned **Whisper** on noisy emergency call recordings for real-time transcription; integrated a BART-based summarization module with call spoof detection achieving **78% precision**.
- Built real-time SOS dispatch and responder tracking backend using WebSockets and Maps API, supporting live coordination under low-latency constraints.
- Selected from **200+ startups** by IITM NIRMAAN; led roadmap planning, agile sprint execution, and MVP delivery across a 5-member cross-functional team.

SmartFan – AI Brand Intelligence & Share-of-Voice Tracker

Personal Project

SOLO DEVELOPER

Aug. 2025

- Built automated brand sentiment and share-of-voice tracking pipeline ingesting data from Twitter, YouTube, and Google APIs via n8n workflow orchestration.
- Integrated **Gemini 2.5 Flash** for multimodal entity tagging and sentiment scoring; achieved **87% F1** on a manually labeled product sentiment evaluation set.
- Designed cross-platform trend dashboards comparing brand visibility across media channels; architecture directly informed GEO retrieval pipeline work at Gravton Labs.

RRT Path Planning for 2D Robot Navigation

SOLO DEVELOPER

Dec. 2024 – Jan. 2025

- Implemented and benchmarked **4 RRT variants** (Basic, Goal-biased, Path-smoothed, and Spacing-aware) with custom collision detection, obstacle avoidance, and path optimization.
- Simulated dynamic environments with narrow passages and moving obstacles in Python; built visualization tools to analyze planner behavior and pathfinding performance.
- Achieved **98% success rate** across **1000+ runs**; reduced path length by **27%** through path smoothing and improved runtime by **41%** compared to baseline RRT.

Extracurricular Activity

- Edited 5+ IITM TV videos using Premiere Pro; content gained 6K+ views and boosted engagement by 23%.
- Reached out to 200+ alumni founders via Tracxn to secure internships and job leads for the graduating batch.
- Mentored 10+ students via IITM Shiksha Prayas; improved fluency and confidence through weekly sessions.